1) a)
2×2

1) b)
2×1

1) c)
1×3

1) d)
2×3

1) e)
3×3

1) f)
1×2

1) g)
3×1

1) h)
3×2

1) i)
5×6

2) a)
$\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}$

2) b)
$\begin{pmatrix} a \\ b \\ c \end{pmatrix}$

2) c)

$$\begin{pmatrix} a & b \\ c & d \\ e & f \\ g & h \end{pmatrix}$$

2) d)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \end{pmatrix}$$

2) e)

$$(a \quad b \quad c \quad d)$$

2) f)

$$\begin{pmatrix} a & b & c \\ d & e & f \\ g & h & i \\ j & k & l \\ m & n & o \end{pmatrix}$$

2) g)

$$\begin{pmatrix} a & b & c & d & e \\ f & g & h & i & j \end{pmatrix}$$

3) a)

$$\begin{pmatrix} 7 & 5 \\ 0 & 9 \end{pmatrix}$$

3) b)

$$\begin{pmatrix} 2 & 9 \\ -3 & -10 \end{pmatrix}$$

3) c)

$$\begin{pmatrix} 11 & -1 & -3 \\ -7 & -10 & 6 \end{pmatrix}$$

3) d)

$$\begin{pmatrix} -9 & 14 \\ 10 & 2 \\ 10 & -4 \end{pmatrix}$$

4) a)

$$\begin{pmatrix} 2 & -6 & -18 \\ 10 & 4 & 8 \end{pmatrix}$$

4) b)

$$\begin{pmatrix} -15 & -10 & 5 \\ 30 & -25 & -10 \\ 10 & -15 & 60 \end{pmatrix}$$

5) a)

$$(5 \quad 7) + 2(-3 \quad 4) = (-1 \quad 15)$$

5) b)

$$3\begin{pmatrix} 3 & -2 \\ 1 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 2 \\ 9 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -8 \\ -6 & 11 \end{pmatrix}$$

5) c)

$$\frac{3}{2}\begin{pmatrix} 8 & 4 \\ 20 & 3 \\ -3 & 2 \end{pmatrix} - 4\begin{pmatrix} -2 & 3 \\ -1 & 2 \\ 8 & \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 20 & -6 \\ 34 & -\frac{7}{2} \\ -\frac{73}{2} & 1 \end{pmatrix}$$

6) a)

$$\begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} + 2\begin{pmatrix} 2 & 0 \\ 1 & -2 \end{pmatrix} + 3\begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} = \begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} + \begin{pmatrix} 4 & 0 \\ 2 & -4 \end{pmatrix} + \begin{pmatrix} 12 & 60 \\ 27 & -33 \end{pmatrix}$$
$$= \begin{pmatrix} 23 & 62 \\ 28 & -34 \end{pmatrix}$$

6) b)

$$\begin{pmatrix} 2 & 0 \\ 1 & -2 \end{pmatrix} - \begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} - \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} = \begin{pmatrix} -9 & -22 \\ -7 & 6 \end{pmatrix}$$

6) c)

$$\begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} - 2\left(\begin{pmatrix} 7 & 2 \\ -1 & 3 \end{pmatrix} + \begin{pmatrix} 2 & 0 \\ 1 & -2 \end{pmatrix}\right) = \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} - 2\begin{pmatrix} 9 & 2 \\ 0 & 1 \end{pmatrix}$$
$$= \begin{pmatrix} 4 & 20 \\ 9 & -11 \end{pmatrix} - \begin{pmatrix} 18 & 4 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} -14 & 16 \\ 9 & -13 \end{pmatrix}$$

7)

$\begin{pmatrix} a & 2 \\ 3 & b \end{pmatrix} + \begin{pmatrix} b & 4 \\ 2 & 2a \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 5 & 10 \end{pmatrix} \Rightarrow \begin{pmatrix} a+b & 6 \\ 5 & 2a+b \end{pmatrix} = \begin{pmatrix} 9 & 6 \\ 5 & 10 \end{pmatrix}$

$a + b = 9$ (1)

$2a + b = 10$ (2)

$\therefore a = 1$ and $b = 8$

8)

$\begin{pmatrix} 4 & k \\ k & m \\ 6 & k \end{pmatrix} - \begin{pmatrix} 2m & 3k \\ m & 5k \\ 2 & m \end{pmatrix} = \begin{pmatrix} -4m & 6m \\ -4m & 16m \\ -2m & -4m \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 4-2m & -2k \\ k-m & m-5k \\ 4 & k-m \end{pmatrix} = \begin{pmatrix} -4m & 6m \\ -4m & 16m \\ -2m & -4m \end{pmatrix}$

$4 - 2m = -4m \Rightarrow 4 = -2m \Rightarrow m = -2$

$-2k = 6m \Rightarrow -2k = -12 \Rightarrow k = 6$

9)

$\begin{pmatrix} 2 & 5q & 8 \\ 3p & p & p \end{pmatrix} + q\begin{pmatrix} 2q & 4 & p \\ q & 3q & 2 \end{pmatrix} = \begin{pmatrix} 4 & -p & q \\ 28 & 12 & 7 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2 & 5q & 8 \\ 3p & p & p \end{pmatrix} + \begin{pmatrix} 2q^2 & 4q & pq \\ q^2 & 3q^2 & 2q \end{pmatrix} = \begin{pmatrix} 4 & -p & q \\ 28 & 12 & 7 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2+2q^2 & 9q & 8+pq \\ 3p+q^2 & p+3q^2 & p+2q \end{pmatrix} = \begin{pmatrix} 4 & -p & q \\ 28 & 12 & 7 \end{pmatrix}$

$9q = -p \Rightarrow p + 9q = 0$ (1)

$p + 2q = 7$ (2)

$\therefore p = 9$ and $q = -1$

10)

Matrices $\mathbf{A}$ and $\mathbf{B}$ have different dimensions (2×2 and 2×3 respectively), and so we cannot find $\mathbf{A} + \mathbf{B}$

11) a)

$\mathbf{A} + \mathbf{B} = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} + \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$

$\mathbf{B} + \mathbf{A} = \begin{pmatrix} 3 & 3 \\ 3 & 3 \end{pmatrix} + \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$

As $\mathbf{A} + \mathbf{B} = \mathbf{B} + \mathbf{A}$, this shows that the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under addition.

11) b)

$\mathbf{A}-\mathbf{B}=\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}-\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}=\begin{pmatrix}-2 & -2\\ -2 & -2\end{pmatrix}$

$\mathbf{B}-\mathbf{A}=\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}-\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}=\begin{pmatrix}2 & 2\\ 2 & 2\end{pmatrix}$

As $\mathbf{A}-\mathbf{B}\neq\mathbf{B}-\mathbf{A}$, this shows that in general matrices are not commutative under subtraction.

12) a)

$\mathbf{A}+(\mathbf{B}+\mathbf{C})=\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}+\left(\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}+\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}\right)=\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}+\begin{pmatrix}8 & 8\\ 8 & 8\end{pmatrix}=\begin{pmatrix}9 & 9\\ 9 & 9\end{pmatrix}$

$(\mathbf{A}+\mathbf{B})+\mathbf{C}=\left(\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}+\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}\right)+\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}=\begin{pmatrix}4 & 4\\ 4 & 4\end{pmatrix}+\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}=\begin{pmatrix}9 & 9\\ 9 & 9\end{pmatrix}$

As $\mathbf{A}+(\mathbf{B}+\mathbf{C})=(\mathbf{A}+\mathbf{B})+\mathbf{C}$, this shows that the matrices **A, B** and **C** are associative under addition.

12) b)

$\mathbf{A}-(\mathbf{B}-\mathbf{C})=\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}-\left(\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}-\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}\right)=\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}-\begin{pmatrix}-2 & -2\\ -2 & -2\end{pmatrix}=\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}$

$(\mathbf{A}-\mathbf{B})-\mathbf{C}=\left(\begin{pmatrix}1 & 1\\ 1 & 1\end{pmatrix}-\begin{pmatrix}3 & 3\\ 3 & 3\end{pmatrix}\right)-\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}=\begin{pmatrix}-2 & -2\\ -2 & -2\end{pmatrix}-\begin{pmatrix}5 & 5\\ 5 & 5\end{pmatrix}=\begin{pmatrix}-7 & -7\\ -7 & -7\end{pmatrix}$

As $\mathbf{A}-(\mathbf{B}-\mathbf{C})\neq(\mathbf{A}-\mathbf{B})-\mathbf{C}$, this shows that in general matrices are not associative under subtraction.

13)

$\mathbf{A}-\mathbf{B}=\begin{pmatrix}2(x+1) & 3(1-x)\\ -3(2-x) & 4(5-x)\end{pmatrix}-\begin{pmatrix}8x+1 & 5x-2\\ 2-x & 3+x\end{pmatrix}$

$=\begin{pmatrix}2x+2 & 3-3x\\ -6+3x & 20-4x\end{pmatrix}-\begin{pmatrix}8x+1 & 5x-2\\ 2-x & 3+x\end{pmatrix}=\begin{pmatrix}1-6x & 5-8x\\ 4x-8 & 17-5x\end{pmatrix}$

14)

$\mathbf{A}-\mathbf{B}=\begin{pmatrix}7k+3 & 3+k\\ 5k+1 & 9-6k\end{pmatrix}-\begin{pmatrix}4k-3 & 13+6k\\ 5+7k & 1-10k\end{pmatrix}=\begin{pmatrix}3k+6 & -10-5k\\ -2k-4 & 8+4k\end{pmatrix}$

$=\begin{pmatrix}3(k+2) & -5(k+2)\\ -2(k+2) & 4(k+2)\end{pmatrix}=(k+2)\begin{pmatrix}3 & -5\\ -2 & 4\end{pmatrix}$

So $\mathrm{f}(k)=k+2$

15) a)

$\mathbf{I}=\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}$

15) b)

$3\mathbf{I} = \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}$

15) c)

$\mathbf{0} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

16) a)

$\mathbf{I} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

16) b)

$-2\mathbf{I} = \begin{pmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{pmatrix}$

16) c)

$\mathbf{0} = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$

17) a)

$\mathbf{0} - \mathbf{A} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix} - \begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} = \begin{pmatrix} -3 & 7 \\ 2 & 5 \end{pmatrix}$

17) b)

$\mathbf{A} + \mathbf{I} = \begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} + \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -7 \\ -2 & -4 \end{pmatrix}$

17) c)

$2\mathbf{I} - 3\mathbf{A} = 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 3\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} - \begin{pmatrix} 9 & -21 \\ -6 & -15 \end{pmatrix} = \begin{pmatrix} -7 & 21 \\ 6 & 17 \end{pmatrix}$

17) d)

$\mathbf{I} - (2\mathbf{A} + \mathbf{0}) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 2\left(\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix} + \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - 2\begin{pmatrix} 3 & -7 \\ -2 & -5 \end{pmatrix}$

$= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} - \begin{pmatrix} 6 & -14 \\ -4 & -10 \end{pmatrix} = \begin{pmatrix} -5 & 14 \\ 4 & 11 \end{pmatrix}$

17) e)

$$\mathbf{0}-(\mathbf{I}-3\mathbf{A})=\begin{pmatrix}0&0\\0&0\end{pmatrix}-\left(\begin{pmatrix}1&0\\0&1\end{pmatrix}-3\begin{pmatrix}3&-7\\-2&-5\end{pmatrix}\right)$$

$$=\begin{pmatrix}0&0\\0&0\end{pmatrix}-\left(\begin{pmatrix}1&0\\0&1\end{pmatrix}-\begin{pmatrix}9&-21\\-6&-15\end{pmatrix}\right)=\begin{pmatrix}0&0\\0&0\end{pmatrix}-\begin{pmatrix}-8&21\\6&16\end{pmatrix}=\begin{pmatrix}8&-21\\-6&-16\end{pmatrix}$$

17) f)

$$10\mathbf{A}-(6\mathbf{I}-5\mathbf{A})=10\begin{pmatrix}3&-7\\-2&-5\end{pmatrix}-\left(6\begin{pmatrix}1&0\\0&1\end{pmatrix}-5\begin{pmatrix}3&-7\\-2&-5\end{pmatrix}\right)$$

$$=\begin{pmatrix}30&-70\\-20&-50\end{pmatrix}-\left(\begin{pmatrix}6&0\\0&6\end{pmatrix}-\begin{pmatrix}15&-35\\-10&-25\end{pmatrix}\right)=\begin{pmatrix}30&-70\\-20&-50\end{pmatrix}-\begin{pmatrix}-9&35\\10&31\end{pmatrix}$$

$$=\begin{pmatrix}39&-105\\-30&-81\end{pmatrix}$$

18)

$$4\begin{pmatrix}a&0&0\\0&b&0\\0&0&3\end{pmatrix}-\begin{pmatrix}b&0&0\\0&2c&0\\0&0&a\end{pmatrix}=2\mathbf{I}$$

$$\Rightarrow\begin{pmatrix}4a&0&0\\0&4b&0\\0&0&12\end{pmatrix}-\begin{pmatrix}b&0&0\\0&2c&0\\0&0&a\end{pmatrix}=\begin{pmatrix}2&0&0\\0&2&0\\0&0&2\end{pmatrix}$$

$4a-b=2$ (1)

$4b-2c=2$ (2)

$12-a=2$ (3)

Solve (3) to get $a=10$

Substitute into (1):

$4(10)-b=2\Rightarrow b=38$

Substitute into (2):

$4(38)-2c=2\Rightarrow c=75$

$\therefore a=10,\ b=38,\ c=75$

19)

$2\begin{pmatrix} p & q \\ 3 & r \end{pmatrix} + k\mathbf{I} = 4\begin{pmatrix} 2q & r \\ p & 2p \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2p & 2q \\ 6 & 2r \end{pmatrix} + \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 8q & 4r \\ 4p & 8p \end{pmatrix}$

$2p + k = 8q$ (1)

$2q = 4r \Rightarrow q - 2r = 0$ (2)

$6 = 4p \Rightarrow p = \frac{3}{2}$ (3)

$2r + k = 8p$ (4)

Substitute (3) into (1) and (4):

$2\left(\frac{3}{2}\right) + k = 8q \Rightarrow k = 8q - 3$ (5)

$2r + k = 8\left(\frac{3}{2}\right) \Rightarrow k = 12 - 2r$ (6)

Solving (5) and (6):

$8q - 3 = 12 - 2r \Rightarrow 8q + 2r = 15$ (7)

Solve (2) and (7): $q = \frac{5}{3}$ and $r = \frac{5}{6}$

$\therefore p = \frac{3}{2}$, $q = \frac{5}{3}$ and $r = \frac{5}{6}$

20)

$\mathbf{AB}$ can be calculated as $\mathbf{A}$ has the same number of columns as the number of rows of $\mathbf{B}$ ($\mathbf{A}$ has 2 columns and $\mathbf{B}$ has 2 rows). $\mathbf{BA}$ cannot be calculated as $\mathbf{B}$ has 2 columns and $\mathbf{A}$ has 3 rows, so these do not match.

21)

$\mathbf{AB}$ can be calculated as $\mathbf{A}$ has the same number of columns as the number of rows of $\mathbf{B}$ ($\mathbf{A}$ has 3 columns and $\mathbf{B}$ has 3 rows). $\mathbf{BA}$ cannot be calculated as $\mathbf{B}$ has 4 columns and $\mathbf{A}$ has 3 rows, so these do not match.

22)

$\mathbf{A}$ is a square matrix and so $\mathbf{A}^2$ can be calculated ($\mathbf{A}$ has 3 columns and 3 rows, so these match). $\mathbf{B}$ is not a square matrix and so $\mathbf{B}^2$ cannot be calculated ($\mathbf{B}$ has 3 columns and 2 rows, so these do not match).

23) a)

$(2 \times 2) \times (2 \times 4) = 2 \times 4$

23) b)

$(2 \times 4) \times (3 \times 4)$ cannot be calculated

23) c)
$(3 \times 2) \times (2 \times 2) = 3 \times 2$

23) d)
$(3 \times 2) \times (3 \times 1)$ cannot be calculated

23) e)
$(3 \times 1) \times (3 \times 2)$ cannot be calculated

23) f)
$(4 \times 2) \times (2 \times 4) = 4 \times 4$

23) g)
$(3 \times 4) \times (4 \times 2) = 3 \times 2$

23) h)
$(2 \times 4) \times (2 \times 2)$ cannot be calculated

24) a)

$$\begin{pmatrix} 2 & 3 & -4 \end{pmatrix} \begin{pmatrix} 2 \\ -1 \\ 2 \end{pmatrix} = (2 \times 2 + 3 \times (-1) + (-4) \times 2) = (-7)$$

24) b)

$$\begin{pmatrix} -3 & 1 & -2 & 4 \end{pmatrix} \begin{pmatrix} 5 \\ -4 \\ 3 \\ 1 \end{pmatrix} = \big((-3) \times 5 + 1 \times (-4) + (-2) \times 3 + 4 \times 1\big) = (-21)$$

24) c)

$$\begin{pmatrix} 4 & 3 \\ -1 & 2 \\ 5 & -6 \end{pmatrix} \begin{pmatrix} 4 \\ 2 \end{pmatrix} = \begin{pmatrix} 4 \times 4 + 3 \times 2 \\ (-1) \times 4 + 2 \times 2 \\ 5 \times 4 + (-6) \times 2 \end{pmatrix} = \begin{pmatrix} 22 \\ 0 \\ 8 \end{pmatrix}$$

24) d)

$$\begin{pmatrix} 3 & 1 \\ 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 1 & 5 \end{pmatrix} = \begin{pmatrix} 3 \times 2 + 1 \times 1 & 3 \times 0 + 1 \times 5 \\ 0 \times 2 + 1 \times 1 & 0 \times 0 + 1 \times 5 \\ 1 \times 2 + 0 \times 1 & 1 \times 0 + 0 \times 5 \end{pmatrix} = \begin{pmatrix} 7 & 5 \\ 1 & 5 \\ 2 & 0 \end{pmatrix}$$

24) e)

$$\begin{pmatrix}5 & 1 & 1\\ 1 & 2 & 1\end{pmatrix}\begin{pmatrix}3 & 2\\ 1 & 2\\ 1 & 1\end{pmatrix}=\begin{pmatrix}5\times3+1\times1+1\times1 & 5\times2+1\times2+1\times1\\ 1\times3+2\times1+1\times1 & 1\times2+2\times2+1\times1\end{pmatrix}=\begin{pmatrix}17 & 13\\ 6 & 7\end{pmatrix}$$

24) f)

$$\begin{pmatrix}3 & 4 & 1\\ 2 & 3 & 0\\ 0 & 3 & 5\end{pmatrix}\begin{pmatrix}1\\ 2\\ 6\end{pmatrix}=\begin{pmatrix}3\times1+4\times2+1\times6\\ 2\times1+3\times2+0\times6\\ 0\times1+3\times2+5\times6\end{pmatrix}=\begin{pmatrix}17\\ 8\\ 36\end{pmatrix}$$

24) g)

$$\begin{pmatrix}3 & 2 & -5\end{pmatrix}\begin{pmatrix}-1 & -2\\ 2 & 6\\ 9 & 0\end{pmatrix}=\begin{pmatrix}3\times-1+2\times2-5\times9 & 3\times-2+2\times6-5\times0\end{pmatrix}$$

$$=\begin{pmatrix}-44 & 6\end{pmatrix}$$

24) h)

$$\begin{pmatrix}4 & 2\\ -3 & -1\end{pmatrix}\begin{pmatrix}5 & 2 & -9\\ -3 & 1 & 10\end{pmatrix}=\begin{pmatrix}4\times5+2\times-3 & 4\times2+2\times1 & 4\times-9+2\times10\\ -3\times5-1\times-3 & -3\times2-1\times1 & -3\times-9-1\times10\end{pmatrix}$$

$$=\begin{pmatrix}14 & 10 & -16\\ -12 & -7 & 17\end{pmatrix}$$

24) i)

$$\begin{pmatrix}4\\ -5\\ 2\end{pmatrix}\begin{pmatrix}1 & -2 & 3 & -8\end{pmatrix}=\begin{pmatrix}4\times1 & 4\times-2 & 4\times3 & 4\times-8\\ -5\times1 & -5\times-2 & -5\times3 & -5\times-8\\ 2\times1 & 2\times-2 & 2\times3 & 2\times-8\end{pmatrix}$$

$$=\begin{pmatrix}4 & -8 & 12 & -32\\ -5 & 10 & -15 & 40\\ 2 & -4 & 6 & -16\end{pmatrix}$$

24) j)

$$\begin{pmatrix}a\\ b\\ c\end{pmatrix}\begin{pmatrix}b & c & a\end{pmatrix}=\begin{pmatrix}ab & ac & a^2\\ b^2 & bc & ab\\ bc & c^2 & ac\end{pmatrix}$$

25)

$$\mathbf{ABA}=\begin{pmatrix}5 & 2 & 1\end{pmatrix}\begin{pmatrix}2\\ -3\\ 4\end{pmatrix}\begin{pmatrix}5 & 2 & 1\end{pmatrix}=\begin{pmatrix}5 & 2 & 1\end{pmatrix}\begin{pmatrix}10 & 4 & 2\\ -15 & -6 & -3\\ 20 & 8 & 4\end{pmatrix}=\begin{pmatrix}40 & 16 & 8\end{pmatrix}$$

26) a)

$$\begin{pmatrix}1 & 0\\ 1 & 1\end{pmatrix}\begin{pmatrix}2 & 2\\ 1 & 1\end{pmatrix}=\begin{pmatrix}1\times2+0\times1 & 1\times2+0\times1\\ 1\times2+1\times1 & 1\times2+1\times1\end{pmatrix}=\begin{pmatrix}2 & 2\\ 3 & 3\end{pmatrix}$$

26) b)

$$\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}^2 = \begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix}\begin{pmatrix}1 & 0\\ 2 & 1\end{pmatrix} = \begin{pmatrix}1\times 1+0\times 2 & 1\times 0+0\times 1\\ 2\times 1+1\times 2 & 2\times 0+1\times 1\end{pmatrix} = \begin{pmatrix}1 & 0\\ 4 & 1\end{pmatrix}$$

26) c)

$$\begin{pmatrix}3 & 1\\ 2 & 5\end{pmatrix}\begin{pmatrix}3 & 2\\ 1 & 5\end{pmatrix} = \begin{pmatrix}3\times 3+1\times 1 & 3\times 2+1\times 5\\ 2\times 3+5\times 1 & 2\times 2+5\times 5\end{pmatrix} = \begin{pmatrix}10 & 11\\ 11 & 29\end{pmatrix}$$

26) d)

$$\begin{pmatrix}-1 & 2\\ -3 & 6\end{pmatrix}\begin{pmatrix}1 & 5\\ -4 & 2\end{pmatrix} = \begin{pmatrix}-1\times 1+2\times -4 & -1\times 5+2\times 2\\ -3\times 1+6\times -4 & -3\times 5+6\times 2\end{pmatrix} = \begin{pmatrix}-9 & -1\\ -27 & -3\end{pmatrix}$$

26) e)

$$\begin{pmatrix}8 & -3\\ -4 & 2\end{pmatrix}\begin{pmatrix}-6 & 5\\ 3 & -5\end{pmatrix} = \begin{pmatrix}8\times -6-3\times 3 & 8\times 5-3\times -5\\ -4\times -6+2\times 3 & -4\times 5+2\times -5\end{pmatrix} = \begin{pmatrix}-57 & 55\\ 30 & -30\end{pmatrix}$$

26) f)

$$\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}^3 = \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix} = \begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}\begin{pmatrix}4 & 0\\ 0 & 4\end{pmatrix} = \begin{pmatrix}8 & 0\\ 0 & 8\end{pmatrix}$$

Alternative Method:

$$\begin{pmatrix}2 & 0\\ 0 & 2\end{pmatrix}^3 = (2\mathbf{I})^3 = 2^3\mathbf{I}^3 = 8\mathbf{I} = \begin{pmatrix}8 & 0\\ 0 & 8\end{pmatrix}$$

26) g)

$$\begin{pmatrix}2 & 1\\ 0 & 2\end{pmatrix}^3 = \begin{pmatrix}2 & 1\\ 0 & 2\end{pmatrix}\begin{pmatrix}2 & 1\\ 0 & 2\end{pmatrix}\begin{pmatrix}2 & 1\\ 0 & 2\end{pmatrix} = \begin{pmatrix}2 & 1\\ 0 & 2\end{pmatrix}\begin{pmatrix}4 & 4\\ 0 & 4\end{pmatrix} = \begin{pmatrix}8 & 12\\ 0 & 8\end{pmatrix}$$

26) h)

$$\begin{pmatrix}\cos\theta & -\sin\theta\\ \sin\theta & \cos\theta\end{pmatrix}\begin{pmatrix}\cos\theta\\ \sin\theta\end{pmatrix} = \begin{pmatrix}\cos^2\theta-\sin^2\theta\\ \sin\theta\cos\theta+\sin\theta\cos\theta\end{pmatrix} = \begin{pmatrix}\cos^2\theta-\sin^2\theta\\ 2\sin\theta\cos\theta\end{pmatrix}$$

This is equal to $\begin{pmatrix}\cos 2\theta\\ \sin 2\theta\end{pmatrix}$ using Double Angle Formulae met in A-Level Maths.

27)

$$\mathbf{A}^2-\mathbf{B}^2 = \begin{pmatrix}1 & 1\\ 3 & 2\end{pmatrix}^2 - \begin{pmatrix}2 & 3\\ 2 & 5\end{pmatrix}^2 = \begin{pmatrix}1 & 1\\ 3 & 2\end{pmatrix}\begin{pmatrix}1 & 1\\ 3 & 2\end{pmatrix} - \begin{pmatrix}2 & 3\\ 2 & 5\end{pmatrix}\begin{pmatrix}2 & 3\\ 2 & 5\end{pmatrix}$$

$$= \begin{pmatrix}4 & 3\\ 9 & 7\end{pmatrix} - \begin{pmatrix}10 & 21\\ 14 & 31\end{pmatrix} = \begin{pmatrix}-6 & -18\\ -5 & -24\end{pmatrix}$$

28)

$$(\mathbf{A}-\mathbf{I})^4=\left(\begin{pmatrix}1 & -3\\ 8 & 1\end{pmatrix}-\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}\right)^4=\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}^4=\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}\begin{pmatrix}0 & -3\\ 8 & 0\end{pmatrix}$$

$$=\begin{pmatrix}-24 & 0\\ 0 & -24\end{pmatrix}\begin{pmatrix}-24 & 0\\ 0 & -24\end{pmatrix}=\begin{pmatrix}576 & 0\\ 0 & 576\end{pmatrix}=576\begin{pmatrix}1 & 0\\ 0 & 1\end{pmatrix}=576\mathbf{I}$$

So $k=576$

29)

$$\begin{pmatrix}k & 0 & 1\\ 1 & k & 2\end{pmatrix}\begin{pmatrix}k & 1\\ 2 & k\\ 1 & k\end{pmatrix}=\begin{pmatrix}1-2\mathrm{i} & 2-2\mathrm{i}\\ 5-3\mathrm{i} & 3-4\mathrm{i}\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}k\times k+0\times 2+1\times 1 & k\times 1+0\times k+1\times k\\ 1\times k+k\times 2+2\times 1 & 1\times 1+k\times k+2\times k\end{pmatrix}=\begin{pmatrix}1-2\mathrm{i} & 2-2\mathrm{i}\\ 5-3\mathrm{i} & 3-4\mathrm{i}\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}k^2+1 & 2k\\ 3k+2 & k^2+2k+1\end{pmatrix}=\begin{pmatrix}1-2\mathrm{i} & 2-2\mathrm{i}\\ 5-3\mathrm{i} & 3-4\mathrm{i}\end{pmatrix}$$

$2k=2-2\mathrm{i}\Rightarrow k=1-\mathrm{i}$

Check the other elements:

$(1-\mathrm{i})^2+1=(1-2\mathrm{i}-1)+1=1-2\mathrm{i}$ as required

$3(1-\mathrm{i})+2=3-3\mathrm{i}+2=5-3\mathrm{i}$ as required

$(1-\mathrm{i})^2+2(1-\mathrm{i})+1=(1-2\mathrm{i}-1)+2-2\mathrm{i}+1=3-4\mathrm{i}$ as required

$\therefore k=1-\mathrm{i}$

30) a)

$$\mathbf{AB}=\begin{pmatrix}4 & -\mathrm{i}\\ 3 & 2\mathrm{i}\end{pmatrix}\begin{pmatrix}k & -3\mathrm{i}\\ 4\mathrm{i} & -2\end{pmatrix}=\begin{pmatrix}4k+4 & -10\mathrm{i}\\ 3k-8 & -13\mathrm{i}\end{pmatrix}$$

30) b)

$$\mathbf{BA}=\begin{pmatrix}k & -3\mathrm{i}\\ 4\mathrm{i} & -2\end{pmatrix}\begin{pmatrix}4 & -\mathrm{i}\\ 3 & 2\mathrm{i}\end{pmatrix}=\begin{pmatrix}4k-9\mathrm{i} & 6-k\mathrm{i}\\ -6+16\mathrm{i} & 4-4\mathrm{i}\end{pmatrix}$$

30) c)

If $\mathbf{AB}=\mathbf{BA}$ then $4k+4=4k-9\mathrm{i}\Rightarrow 4=-9\mathrm{i}$, which is not true

$\therefore$ there is no value of k such that $\mathbf{AB}=\mathbf{BA}$

31)

$$\mathbf{AC} = \begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix}\begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -11 & 23 \\ -52 & 85 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} 5 & 6 \\ -9 & 20 \end{pmatrix}\begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix} = \begin{pmatrix} 28 & -59 & 75 \\ 42 & -171 & 173 \end{pmatrix}$$

$$\mathbf{CA} = \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 2 & -1 & 3 \\ 3 & -9 & 10 \end{pmatrix} = \begin{pmatrix} 17 & -61 & 64 \\ -10 & 50 & -48 \\ 1 & -8 & 7 \end{pmatrix}$$

$$\mathbf{CB} = \begin{pmatrix} -2 & 7 \\ 4 & -6 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 5 & 6 \\ -9 & 20 \end{pmatrix} = \begin{pmatrix} -73 & 128 \\ 74 & -96 \\ -14 & 14 \end{pmatrix}$$

32) a)

$$\begin{pmatrix} a & 2 \\ 2a & 1 \end{pmatrix}\begin{pmatrix} 3 & a \\ a & 1 \end{pmatrix} = \begin{pmatrix} 3a + 2a & a^2 + 2 \\ 6a + a & 2a^2 + 1 \end{pmatrix} = \begin{pmatrix} 5a & a^2 + 2 \\ 7a & 2a^2 + 1 \end{pmatrix}$$

32) b)

$$\begin{pmatrix} 2b & b \\ 1 & 2 \end{pmatrix}\begin{pmatrix} 4 & 3b \\ 2 & 2b \end{pmatrix} = \begin{pmatrix} 8b + 2b & 6b^2 + 2b^2 \\ 8 & 3b + 4b \end{pmatrix} = \begin{pmatrix} 10b & 8b^2 \\ 8 & 7b \end{pmatrix}$$

33)

$$\mathbf{AB} = \begin{pmatrix} -2 & k & 0 \\ 1 & k & 1 \end{pmatrix}\begin{pmatrix} 1 & 2 \\ 2 & k-1 \\ -1 & 2k \end{pmatrix}$$

$$= \begin{pmatrix} -2 + 2k + 0 & -4 + k(k-1) + 0 \\ 1 + 2k - 1 & 2 + k(k-1) + 2k \end{pmatrix}$$

$$= \begin{pmatrix} 2k - 2 & k^2 - k - 4 \\ 2k & k^2 + k + 2 \end{pmatrix}$$

As $\mathbf{AB}$ is symmetric, we must have that $2k = k^2 - k - 4 \Rightarrow 0 = k^2 - 3k - 4$

$\Rightarrow 0 = (k-4)(k+1) \Rightarrow k = -1$ or $k = 4$

When $k = -1$, $\mathbf{AB} = \begin{pmatrix} -4 & -2 \\ -2 & 2 \end{pmatrix}$

When $k = 4$, $\mathbf{AB} = \begin{pmatrix} 6 & 8 \\ 8 & 22 \end{pmatrix}$

34) a)

$$\mathbf{AB} = \begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix}\begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 4 \\ 0 & 4 & 5 \end{pmatrix} = \begin{pmatrix} 2 & 11 & 11 \\ 3 & 0 & 12 \\ 8 & 32 & 29 \end{pmatrix}$$

34) b)

$$\mathbf{BA} = \begin{pmatrix} 2 & 3 & 1 \\ 1 & 0 & 4 \\ 0 & 4 & 5 \end{pmatrix}\begin{pmatrix} 1 & 0 & 2 \\ 0 & 3 & 0 \\ 4 & 0 & 5 \end{pmatrix} = \begin{pmatrix} 6 & 9 & 9 \\ 17 & 0 & 22 \\ 20 & 12 & 25 \end{pmatrix}$$

35) a)

$$\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}^3 = \begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix}$$

$$= \begin{pmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & c^2 \end{pmatrix}\begin{pmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{pmatrix} = \begin{pmatrix} a^3 & 0 & 0 \\ 0 & b^3 & 0 \\ 0 & 0 & c^3 \end{pmatrix}$$

35) b)

$$\mathbf{A}^3 = \begin{pmatrix} \sin\theta & 0 & 0 \\ 0 & -\cos\theta & 0 \\ 0 & 0 & -\sin\theta \end{pmatrix}^3 = \begin{pmatrix} (\sin\theta)^3 & 0 & 0 \\ 0 & (-\cos\theta)^3 & 0 \\ 0 & 0 & (-\sin\theta)^3 \end{pmatrix}$$

$$= \begin{pmatrix} \sin^3\theta & 0 & 0 \\ 0 & -\cos^3\theta & 0 \\ 0 & 0 & -\sin^3\theta \end{pmatrix}$$

36)

Let $\mathbf{A} = \begin{pmatrix} a \\ b \end{pmatrix}$

$$\begin{pmatrix} 2 & 3 \\ -3 & 4 \end{pmatrix}\begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} 12 \\ -69 \end{pmatrix} \Rightarrow \begin{pmatrix} 2a+3b \\ -3a+4b \end{pmatrix} = \begin{pmatrix} 12 \\ -69 \end{pmatrix}$$

$2a + 3b = 12$ (1)

$-3a + 4b = -69$ (2)

$\therefore a = 15$ and $b = -6$

$\therefore \mathbf{A} = \begin{pmatrix} 15 \\ -6 \end{pmatrix}$

37)

Let $\mathbf{A} = \begin{pmatrix} a & b & c \end{pmatrix}$

$\begin{pmatrix} a & b & c \end{pmatrix}\begin{pmatrix} -4 & 1 \\ 0 & 0 \\ 4 & 5 \end{pmatrix} = \begin{pmatrix} 32 & 24 \end{pmatrix} \Rightarrow \begin{pmatrix} -4a + 4c & a + 5c \end{pmatrix} = \begin{pmatrix} 32 & 24 \end{pmatrix}$

$-4a + 4c = 32$ (1)

$a + 5c = 24$ (2)

$\therefore a = -\frac{8}{3}$ and $c = \frac{16}{3}$

$\therefore \mathbf{A} = \begin{pmatrix} -\frac{8}{3} & b & \frac{16}{3} \end{pmatrix}$ where $b \in \mathbb{R}$

So a possible matrix could be $\mathbf{A} = \begin{pmatrix} -\frac{8}{3} & 1 & \frac{16}{3} \end{pmatrix}$

38)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\begin{pmatrix} 2 & -3 \\ 0 & 6 \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} 13 & 1 \\ -6 & -18 \end{pmatrix} \Rightarrow \begin{pmatrix} 2a - 3c & 2b - 3d \\ 6c & 6d \end{pmatrix} = \begin{pmatrix} 13 & 1 \\ -6 & -18 \end{pmatrix}$

$6c = -6 \Rightarrow c = -1$

$6d = -18 \Rightarrow d = -3$

$2a - 3(-1) = 13 \Rightarrow a = 5$

$2b - 3(-3) = 1 \Rightarrow b = -4$

$\therefore \mathbf{A} = \begin{pmatrix} 5 & -4 \\ -1 & -3 \end{pmatrix}$

39)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} -5 & 2 & -4 \\ 10 & 2 & 16 \end{pmatrix} = \begin{pmatrix} -15 & 6 & -12 \\ 15 & 6 & 28 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} -5a + 10b & 2a + 2b & -4a + 16b \\ -5c + 10d & 2c + 2d & -4c + 16d \end{pmatrix} = \begin{pmatrix} -15 & 6 & -12 \\ 15 & 6 & 28 \end{pmatrix}$

$-5a + 10b = -15$ (1)

$2a + 2b = 6$ (2)

$-4a + 16b = -12$ (3)

$\therefore a = 3$ and $b = 0$

$-5c + 10d = 15$ (4)

$2c + 2d = 6$ (5)

$-4c + 16d = 28$ (6)

$\therefore c = 1$ and $d = 2$

$\therefore \mathbf{A} = \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix}$

40)

$$\mathbf{AB} = \begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix}\begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 6\times 2 + 2\times 0 & 6\times 0 + 2\times 2 \\ 3\times 2 + 1\times 0 & 3\times 0 + 1\times 2 \end{pmatrix} = \begin{pmatrix} 12 & 4 \\ 6 & 2 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}\begin{pmatrix} 6 & 2 \\ 3 & 1 \end{pmatrix} = \begin{pmatrix} 2\times 6 + 0\times 3 & 2\times 2 + 0\times 1 \\ 0\times 6 + 2\times 3 & 0\times 2 + 2\times 1 \end{pmatrix} = \begin{pmatrix} 12 & 4 \\ 6 & 2 \end{pmatrix}$$

As $\mathbf{AB} = \mathbf{BA}$, $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

41)

$$\mathbf{AB} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix} = \begin{pmatrix} 1\times 3 + 1\times 3 & 1\times 0 + 1\times 3 \\ 0\times 3 + 1\times 3 & 0\times 0 + 1\times 3 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 3 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 3\times 1 + 0\times 0 & 3\times 1 + 0\times 1 \\ 3\times 1 + 3\times 0 & 3\times 1 + 3\times 1 \end{pmatrix} = \begin{pmatrix} 3 & 3 \\ 3 & 6 \end{pmatrix}$$

As $\mathbf{AB} \neq \mathbf{BA}$, matrix multiplication is not, in general, commutative.

42)

$$\mathbf{A(BC)} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\left(\begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix}\right) = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 15 & 15 \\ 30 & 15 \end{pmatrix} = \begin{pmatrix} 45 & 30 \\ 30 & 15 \end{pmatrix}$$

$$\mathbf{(AB)C} = \left(\begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 3 & 0 \\ 3 & 3 \end{pmatrix}\right)\begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 6 & 3 \\ 3 & 3 \end{pmatrix}\begin{pmatrix} 5 & 5 \\ 5 & 0 \end{pmatrix} = \begin{pmatrix} 45 & 30 \\ 30 & 15 \end{pmatrix}$$

As $\mathbf{A(BC)} = \mathbf{(AB)C}$, $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under multiplication.

43) a)

$$\mathbf{(AB)C} = \left(\begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix}\right)\begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix} = \begin{pmatrix} -2k+3 & 12 \\ -5 & -4 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}$$

$$= \begin{pmatrix} -2k+3 & 4k+30 \\ -5 & -2 \end{pmatrix}$$

$$\mathbf{A(BC)} = \begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix}\left(\begin{pmatrix} -2 & 0 \\ 1 & 4 \end{pmatrix}\begin{pmatrix} 1 & -2 \\ 0 & 3 \end{pmatrix}\right) = \begin{pmatrix} k & 3 \\ 2 & -1 \end{pmatrix}\begin{pmatrix} -2 & 4 \\ 1 & 10 \end{pmatrix}$$

$$= \begin{pmatrix} -2k+3 & 4k+30 \\ -5 & -2 \end{pmatrix}$$

$\therefore \mathbf{(AB)C} = \mathbf{A(BC)}$

43) b)

This shows that $\mathbf{A}$, $\mathbf{B}$ and $\mathbf{C}$ are associative under matrix multiplication

44)

$$\mathbf{(AB)^2} = \left(\begin{pmatrix} 5 & 2 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} -2 & 1 \\ 0 & 4 \end{pmatrix}\right)^2 = \begin{pmatrix} -10 & 13 \\ -2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 74 & -117 \\ 18 & -25 \end{pmatrix}$$

$$\mathbf{A^2B^2} = \begin{pmatrix} 5 & 2 \\ 1 & 0 \end{pmatrix}^2\begin{pmatrix} -2 & 1 \\ 0 & 4 \end{pmatrix}^2 = \begin{pmatrix} 27 & 10 \\ 5 & 2 \end{pmatrix}\begin{pmatrix} 4 & 2 \\ 0 & 16 \end{pmatrix} = \begin{pmatrix} 108 & 214 \\ 20 & 42 \end{pmatrix}$$

$\therefore \mathbf{(AB)^2} \neq \mathbf{A^2B^2}$

45)

$\begin{pmatrix} 5a & -3 \\ 1 & 2 \end{pmatrix}\begin{pmatrix} p \\ q \end{pmatrix} = 4\begin{pmatrix} p \\ q \end{pmatrix} \Rightarrow \begin{pmatrix} 5ap - 3q \\ p + 2q \end{pmatrix} = \begin{pmatrix} 4p \\ 4q \end{pmatrix}$

$5ap - 3q = 4p$ (1)

$p + 2q = 4q$ (2)

(2) $\Rightarrow p = 2q$

(1) $\Rightarrow 5a \times 2q - 3q = 4 \times 2q \Rightarrow 10a - 3 = 8 \Rightarrow a = \frac{11}{10}$

46)

$\begin{pmatrix} a & 1 & 2 \\ 0 & 3 & a \\ a & 2 & 1 \end{pmatrix}^2 = \begin{pmatrix} 15 & 2 & -13 \\ 25 & -1 & -20 \\ 20 & 3 & -19 \end{pmatrix} \Rightarrow \begin{pmatrix} a^2 + 2a & a + 7 & 3a + 2 \\ a^2 & 9 + 2a & 4a \\ a^2 + a & a + 8 & 4a + 1 \end{pmatrix} = \begin{pmatrix} 15 & 2 & -13 \\ 25 & -1 & -20 \\ 20 & 3 & -19 \end{pmatrix}$

$a^2 + 2a = 15 \Rightarrow a = -5$ or $a = 3$

$a^2 = 25 \Rightarrow a = 5$ or $a = -5$

$a^2 + a = 20 \Rightarrow a = -5$ or $a = 4$

$a + 7 = 2 \Rightarrow a = -5$

$9 + 2a = -1 \Rightarrow a = -5$

$a + 8 = 3 \Rightarrow a = -5$

$3a + 2 = -13 \Rightarrow a = -5$

$4a = -20 \Rightarrow a = -5$

$4a + 1 = -19 \Rightarrow a = -5$

$\therefore a = -5$

47)

$\begin{pmatrix} 2 & 2 & -1 \\ 2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}\begin{pmatrix} 9 & 6 & 3 \\ 2 & 0 & x \\ -2 & -6 & 4 \end{pmatrix} = \begin{pmatrix} 9 & 6 & 3 \\ 2 & 0 & x \\ -2 & -6 & 4 \end{pmatrix}\begin{pmatrix} 2 & 2 & -1 \\ 2 & 1 & -2 \\ -2 & -2 & 1 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 24 & 18 & 2x + 2 \\ 24 & 24 & x - 2 \\ -24 & -18 & -2x - 2 \end{pmatrix} = \begin{pmatrix} 24 & 18 & -18 \\ 4 - 2x & 4 - 2x & x - 2 \\ -24 & -18 & 18 \end{pmatrix}$

$2x + 2 = -18 \Rightarrow x = -10$

$4 - 2x = 24 \Rightarrow x = -10$

$\therefore x = -10$

48) a)

$\mathbf{A}^2 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^2 = \begin{pmatrix} 1 & 0 \\ 8 & 1 \end{pmatrix}$

48) b)

$\mathbf{A}^3 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^3 = \begin{pmatrix} 1 & 0 \\ 12 & 1 \end{pmatrix}$

48) c)

$$\mathbf{A}^4 = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^4 = \begin{pmatrix} 1 & 0 \\ 16 & 1 \end{pmatrix}$$

48) d)

$$\mathbf{A}^k = \begin{pmatrix} 1 & 0 \\ 4 & 1 \end{pmatrix}^k = \begin{pmatrix} 1 & 0 \\ 4k & 1 \end{pmatrix}$$

49) a)

$$\mathbf{A}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^2 = \begin{pmatrix} 2 & 2 \\ 2 & 2 \end{pmatrix}$$

49) b)

$$\mathbf{A}^3 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^3 = \begin{pmatrix} 4 & 4 \\ 4 & 4 \end{pmatrix}$$

49) c)

$$\mathbf{A}^4 = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^4 = \begin{pmatrix} 8 & 8 \\ 8 & 8 \end{pmatrix}$$

49) d)

$$\mathbf{A}^k = \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}^k = \begin{pmatrix} 2^{k-1} & 2^{k-1} \\ 2^{k-1} & 2^{k-1} \end{pmatrix} = 2^{k-1}\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}$$

50) a)

$$\mathbf{A}^2 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^2 = \begin{pmatrix} 3 & 2 \\ -2 & -1 \end{pmatrix}$$

50) b)

$$\mathbf{A}^3 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^3 = \begin{pmatrix} 4 & 3 \\ -3 & -2 \end{pmatrix}$$

50) c)

$$\mathbf{A}^4 = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^4 = \begin{pmatrix} 5 & 4 \\ -4 & -3 \end{pmatrix}$$

50) d)

$$\mathbf{A}^k = \begin{pmatrix} 2 & 1 \\ -1 & 0 \end{pmatrix}^k = \begin{pmatrix} k+1 & k \\ -k & 1-k \end{pmatrix}$$

51)

$\begin{pmatrix} 2 & a \\ 2b & 1 \end{pmatrix}\begin{pmatrix} 3a & 4 \\ 2 & 2b \end{pmatrix} = \begin{pmatrix} 48 & 116 \\ 326 & 90 \end{pmatrix} \Rightarrow \begin{pmatrix} 8a & 8+2ab \\ 6ab+2 & 10b \end{pmatrix} = \begin{pmatrix} 48 & 116 \\ 326 & 90 \end{pmatrix}$

$8a = 48 \Rightarrow a = 6$

$10b = 90 \Rightarrow b = 9$

Check: $8 + 2ab = 8 + 2 \times 6 \times 9 = 116$

Check: $6ab + 2 = 6 \times 6 \times 9 + 2 = 326$

$\therefore a = 6$ and $b = 9$

52)

$\begin{pmatrix} 3a & -2 \\ 5a & -3 \end{pmatrix}\begin{pmatrix} 8 & -3a \\ 4a & 1 \end{pmatrix} = \begin{pmatrix} b+2 & 22-2b \\ 2b-4 & -2b-3 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 16a & -9a^2-2 \\ 28a & -15a^2-3 \end{pmatrix} = \begin{pmatrix} b+2 & 22-2b \\ 2b-4 & -2b-3 \end{pmatrix}$

$16a = b + 2 \Rightarrow 16a - b = 2$ (1)

$28a = 2b - 4 \Rightarrow 28a - 2b = -4$ (2)

$\therefore a = 2$ and $b = 30$

$-9a^2 - 2 = 22 - 2b \Rightarrow -9a^2 + 2b = 24$

Check: $-9a^2 + 2b = -9 \times 2^2 + 2 \times 30 = 24$

$-15a^2 - 3 = -2b - 3 \Rightarrow -15a^2 + 2b = 0$

Check: $-15a^2 + 2b = -15 \times 2^2 + 2 \times 30 = 0$

$\therefore a = 2$ and $b = 30$

53)

$\begin{pmatrix} b & 5 \\ 2a & 3 \end{pmatrix}\begin{pmatrix} b & 5 \\ 2a & 3 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \Rightarrow \begin{pmatrix} b^2+10a & 5b+15 \\ 2ab+6a & 10a+9 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$

$10a + 9 = 1 \Rightarrow a = -\frac{4}{5}$

$5b + 15 = 0 \Rightarrow b = -3$

Check: $b^2 + 10a = (-3)^2 + 10 \times -\frac{4}{5} = 1$

Check: $2ab + 6a = 2 \times -\frac{4}{5} \times -3 + 6 \times -\frac{4}{5} = 0$

$\therefore a = -\frac{4}{5}$ and $b = -3$

54)

$\begin{pmatrix} 3b & 4 \\ 5a & 2 \end{pmatrix}^2 = \mathbf{0} \Rightarrow \begin{pmatrix} 20a+9b^2 & 12b+8 \\ 15ab+10a & 20a+4 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$

$20a + 4 = 0 \Rightarrow a = -\frac{1}{5}$

$12b + 8 = 0 \Rightarrow b = -\frac{2}{3}$

55)

$p\begin{pmatrix}2 & 5\\10 & 2\end{pmatrix}-\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}=q\begin{pmatrix}7 & 2\\4 & 7\end{pmatrix}\Rightarrow\begin{pmatrix}2p-1 & 5p\\10p & 2p-1\end{pmatrix}=\begin{pmatrix}7q & 2q\\4q & 7q\end{pmatrix}$

$2p-1=7q\Rightarrow 2p-7q=1$ (1)

$5p=2q\Rightarrow 5p-2q=0$ (2)

$\therefore p=-\frac{2}{31}$ and $q=-\frac{5}{31}$

56)

$p\begin{pmatrix}1 & 10\\50 & 1\end{pmatrix}+q\begin{pmatrix}1 & 3\\15 & 1\end{pmatrix}=5\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}\Rightarrow\begin{pmatrix}p & 10p\\50p & p\end{pmatrix}+\begin{pmatrix}q & 3q\\15q & q\end{pmatrix}=\begin{pmatrix}5 & 0\\0 & 5\end{pmatrix}$

$\Rightarrow\begin{pmatrix}p+q & 10p+3q\\50p+15p & p+q\end{pmatrix}=\begin{pmatrix}5 & 0\\0 & 5\end{pmatrix}$

$p+q=5$ (1)

$10p+3q=0$ (2)

$\therefore p=-\frac{15}{7}$ and $q=\frac{50}{7}$

57)

$\begin{pmatrix}5 & 1\\x & 2\\0 & 1\end{pmatrix}\begin{pmatrix}2y & 3z\\z & y\end{pmatrix}=\begin{pmatrix}38 & 123\\52 & 150\\8 & 3\end{pmatrix}=\begin{pmatrix}10y+z & 15z+y\\2xy+2z & 3xz+2y\\z & y\end{pmatrix}=\begin{pmatrix}38 & 123\\52 & 150\\8 & 3\end{pmatrix}$

$\therefore z=8$ and $y=3$

$2xy+2z=52\Rightarrow 2x\times 3+2\times 8=52\Rightarrow 6x+16=52\Rightarrow x=6$

Check: $10y+z=10\times 3+8=38$

Check: $15z+y=15\times 8+3=123$

Check: $3xz+2y=3\times 6\times 8+2\times 3=150$

$\therefore x=6$, $y=3$ and $z=8$

58)

$\begin{pmatrix}-1 & x & 3 & 4\\x & 3 & y & y\end{pmatrix}\begin{pmatrix}4 & 5\\2 & 3\\x & y\\4 & 2\end{pmatrix}=2\begin{pmatrix}1 & 0\\0 & 1\end{pmatrix}$

$\Rightarrow\begin{pmatrix}5x+12 & 3x+3y+3\\4x+6+xy+4y & 5x+9+y^2+2y\end{pmatrix}=\begin{pmatrix}2 & 0\\0 & 2\end{pmatrix}$

$5x+12=2\Rightarrow x=-2$

$3\times -2+3y+3=0\Rightarrow y=1$

Check: $4x+6+xy+4y=4\times -2+6-2\times 1+4\times 1=0$

Check: $5x+9+y^2+2y=5\times -2+9+1^2+2\times 1=2$

$\therefore x=-2$ and $y=1$

59)

$$\begin{pmatrix} 5 & 3 \\ 18 & 2 \end{pmatrix}\begin{pmatrix} x+3 & 1 \\ x-2 & 10 \end{pmatrix} = \begin{pmatrix} x+3 & 1 \\ x-2 & 10 \end{pmatrix}\begin{pmatrix} 5 & 3 \\ 18 & 2 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 5(x+3)+3(x-2) & 35 \\ 18(x+3)+2(x-2) & 38 \end{pmatrix} = \begin{pmatrix} 5(x+3)+18 & 3(x+3)+2 \\ 5(x-2)+180 & 3(x-2)+20 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 8x+9 & 35 \\ 20x+50 & 38 \end{pmatrix} = \begin{pmatrix} 5x+33 & 3x+11 \\ 5x+170 & 3x+14 \end{pmatrix}$$

$8x + 9 = 5x + 33 \Rightarrow 3x = 24 \Rightarrow x = 8$

$35 = 3x + 11 \Rightarrow 3x = 24 \Rightarrow x = 8$

$20x + 50 = 5x + 170 \Rightarrow 15x = 120 \Rightarrow x = 8$

$38 = 3x + 14 \Rightarrow 3x = 24 \Rightarrow x = 8$

$\therefore x = 8$

60)

$$\begin{pmatrix} k & 0 \\ k+2 & 3-k \end{pmatrix}\begin{pmatrix} 2 & 0 \\ -k & 3 \end{pmatrix} = \begin{pmatrix} 2 & 0 \\ -k & 3 \end{pmatrix}\begin{pmatrix} k & 0 \\ k+2 & 3-k \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 2k & 0 \\ k^2-k+4 & 9-3k \end{pmatrix} = \begin{pmatrix} 2k & 0 \\ -k^2+3k+6 & 9-3k \end{pmatrix}$$

$k^2 - k + 4 = -k^2 + 3k + 6$

$\Rightarrow 2k^2 - 4k - 2 = 0$

$\Rightarrow k^2 - 2k - 1 = 0$

$\Rightarrow (k-1)^2 - 1 - 1 = 0$

$\Rightarrow (k-1)^2 = 2$

$\Rightarrow k = 1 \pm \sqrt{2}$

61)

$$\mathbf{AB} = \begin{pmatrix} 2x^2 & 16x \\ 8x & x \end{pmatrix}\begin{pmatrix} x+1 & 16 \\ 8 & 2-x \end{pmatrix} = \begin{pmatrix} 2x^2(x+1)+128x & 32x^2+16x(2-x) \\ 8x(x+1)+8x & 128x+x(2-x) \end{pmatrix}$$

$$= \begin{pmatrix} 2x^3+2x^2+128x & 16x^2+32x \\ 8x^2+16x & 130x-x^2 \end{pmatrix}$$

$$\mathbf{BA} = \begin{pmatrix} x+1 & 16 \\ 8 & 2-x \end{pmatrix}\begin{pmatrix} 2x^2 & 16x \\ 8x & x \end{pmatrix} = \begin{pmatrix} 2x^2(x+1)+128x & 16x(x+1)+16x \\ 16x^2+8x(2-x) & 128x+x(2-x) \end{pmatrix}$$

$$= \begin{pmatrix} 2x^3+2x^2+128x & 16x^2+32x \\ 8x^2+16x & 130x-x^2 \end{pmatrix}$$

$\mathbf{AB} = \mathbf{BA}$ so the matrices $\mathbf{A}$ and $\mathbf{B}$ are commutative under multiplication.

62)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$4\begin{pmatrix} a & b \\ c & d \end{pmatrix} - 2\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 1 & 2 \\ -3 & 4 \end{pmatrix} \Rightarrow \begin{pmatrix} 4a-2 & 4b \\ 4c & 4d-2 \end{pmatrix} = \begin{pmatrix} a-3b & 2a+4b \\ c-3d & 2c+4d \end{pmatrix}$

$4a - 2 = a - 3b \Rightarrow 3a + 3b = 2$

$4b = 2a + 4b \Rightarrow 2a = 0 \Rightarrow a = 0$

$\therefore 3 \times 0 + 3b = 2 \Rightarrow b = \frac{2}{3}$

$4c = c - 3d \Rightarrow 3c + 3d = 0 \Rightarrow c + d = 0$

$4d - 2 = 2c + 4d \Rightarrow 2c = -2 \Rightarrow c = -1$

$\therefore -1 + d = 0 \Rightarrow d = 1$

$\therefore \mathbf{A} = \begin{pmatrix} 0 & \frac{2}{3} \\ -1 & 1 \end{pmatrix}$

63)

$k\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} a+3 & -2a \\ 6-a & a \end{pmatrix}\begin{pmatrix} a-6 & a-3 \\ 10-a & a-5 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} (a+3)(a-6) - 2a(10-a) & (a+3)(a-3) - 2a(a-5) \\ (6-a)(a-6) + a(10-a) & (6-a)(a-3) + a(a-5) \end{pmatrix}$

$\Rightarrow \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} a^2 - 3a - 18 - 20a + 2a^2 & a^2 - 9 - 2a^2 + 10a \\ -a^2 + 12a - 36 + 10a - a^2 & -a^2 + 9a - 18 + a^2 - 5a \end{pmatrix}$

$\Rightarrow \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = \begin{pmatrix} 3a^2 - 23a - 18 & -a^2 + 10a - 9 \\ -2a^2 + 22a - 36 & 4a - 18 \end{pmatrix}$

$-a^2 + 10a - 9 = 0 \Rightarrow a = 9$ or $a = 1$

$-2a^2 + 22a - 36 = 0 \Rightarrow a = 9$ or $a = 2$

$\therefore a = 9$

$k = 3 \times 9^2 - 23 \times 9 - 18 = 18$

Check: $4 \times 9 - 18 = 18$

$\therefore a = 9$ and $k = 18$

64)

$$\begin{pmatrix}1 & -1 & -3\\ 0 & x & 1\\ 3 & -2 & 1\end{pmatrix}\begin{pmatrix}\frac{4}{3} & 1 & -\frac{5}{3}\\ 2 & x+13 & 3\\ 1 & -8 & 0\end{pmatrix}=\begin{pmatrix}\frac{4}{3} & 1 & -\frac{5}{3}\\ 2 & x+13 & 3\\ 1 & -8 & 0\end{pmatrix}\begin{pmatrix}1 & -1 & -3\\ 0 & x & 1\\ 3 & -2 & 1\end{pmatrix}$$

$$\Rightarrow\begin{pmatrix}-\frac{11}{3} & 12-x & -\frac{14}{3}\\ 2x+1 & x^2+13x-8 & 3x\\ 1 & -2x-31 & -11\end{pmatrix}=\begin{pmatrix}-\frac{11}{3} & x+2 & -\frac{14}{3}\\ 11 & x^2+13x-8 & x+10\\ 1 & -8x-1 & -11\end{pmatrix}$$

$12-x=x+2\Rightarrow 2x=10\Rightarrow x=5$

$2x+1=11\Rightarrow 2x=10\Rightarrow x=5$

$3x=x+10\Rightarrow 2x=10\Rightarrow x=5$

$-2x-31=-8x-1\Rightarrow 6x=30\Rightarrow x=5$

$\therefore x=5$

65)

Let $\mathbf{A}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ and $\mathbf{B}=\begin{pmatrix}e & f\\ g & h\end{pmatrix}$

$$\mathbf{A}+\mathbf{B}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}+\begin{pmatrix}e & f\\ g & h\end{pmatrix}=\begin{pmatrix}a+e & b+f\\ c+g & d+h\end{pmatrix}$$

$$\mathbf{B}+\mathbf{A}=\begin{pmatrix}e & f\\ g & h\end{pmatrix}+\begin{pmatrix}a & b\\ c & d\end{pmatrix}=\begin{pmatrix}e+a & f+b\\ g+c & h+d\end{pmatrix}=\begin{pmatrix}a+e & b+f\\ c+g & d+h\end{pmatrix}$$

So $\mathbf{A}+\mathbf{B}=\mathbf{B}+\mathbf{A}$

$\therefore$ matrix addition is commutative for 2 × 2 matrices

66)

Let $\mathbf{A}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}$ and $\mathbf{B}=\begin{pmatrix}e & f\\ g & h\end{pmatrix}$

$$\mathbf{A}-\mathbf{B}=\begin{pmatrix}a & b\\ c & d\end{pmatrix}-\begin{pmatrix}e & f\\ g & h\end{pmatrix}=\begin{pmatrix}a-e & b-f\\ c-g & d-h\end{pmatrix}$$

$$\mathbf{B}-\mathbf{A}=\begin{pmatrix}e & f\\ g & h\end{pmatrix}-\begin{pmatrix}a & b\\ c & d\end{pmatrix}=\begin{pmatrix}e-a & f-b\\ g-c & h-d\end{pmatrix}$$

In general, $a-e\neq e-a$, $b-f\neq f-b$, $c-g\neq g-c$, $d-h\neq h-d$

So $\mathbf{A}-\mathbf{B}\neq\mathbf{B}-\mathbf{A}$

$\therefore$ matrix subtraction is not commutative for 2 × 2 matrices

67)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$

$\mathbf{AB} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}$

$\mathbf{BA} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}\begin{pmatrix} a & b \\ c & d \end{pmatrix} = \begin{pmatrix} ae+cf & be+df \\ ag+ch & bg+dh \end{pmatrix}$

In general, $ae+bg \neq ae+cf$, $af+bh \neq be+df$, $ce+dg \neq ag+ch$, $cf+dh \neq bg+dh$

So $\mathbf{AB} \neq \mathbf{BA}$

∴ matrix multiplication is not in general commutative for 2 × 2 matrices

68) a)

$\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} 2 & 6 \\ 3 & -7 \end{pmatrix}$

68) b)

$\mathbf{B}^{\mathrm{T}} = \begin{pmatrix} 4 & -9 \\ 2 & 1 \\ 0 & 4 \end{pmatrix}$

68) c)

$\mathbf{C}^{\mathrm{T}} = \begin{pmatrix} 1 & 10 & 3 \\ 2 & -9 & 0 \\ -7 & 2 & 8 \end{pmatrix}$

69)

$\mathbf{A} = \begin{pmatrix} 5 & -8 & 5 \\ 2 & 1 & 3 \end{pmatrix}$, $\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} 5 & 2 \\ -8 & 1 \\ 5 & 3 \end{pmatrix}$, $\mathbf{B} = \begin{pmatrix} 8 & -5 \\ 2 & -1 \end{pmatrix}$, $\mathbf{B}^{\mathrm{T}} = \begin{pmatrix} 8 & 2 \\ -5 & -1 \end{pmatrix}$

$\mathbf{AA}^{\mathrm{T}}$, $\mathbf{A}^{\mathrm{T}}\mathbf{A}$, $\mathbf{A}^{\mathrm{T}}\mathbf{B}$, $\mathbf{A}^{\mathrm{T}}\mathbf{B}^{\mathrm{T}}$, $\mathbf{BA}$, $\mathbf{BB}^{\mathrm{T}}$, $\mathbf{B}^{\mathrm{T}}\mathbf{A}$, $\mathbf{B}^{\mathrm{T}}\mathbf{B}$

70) a)

$\mathbf{AA}^{\mathrm{T}} = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}\begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}\begin{pmatrix} k & 4 \\ 2k & 6k \end{pmatrix} = \begin{pmatrix} 45 & 96 \\ 96 & 340 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 5k^2 & 12k^2+4k \\ 12k^2+4k & 36k^2+16 \end{pmatrix} = \begin{pmatrix} 45 & 96 \\ 96 & 340 \end{pmatrix}$

$5k^2 = 45 \Rightarrow k^2 = 9 \Rightarrow k = \pm 3$

$12k^2 + 4k = 96 \Rightarrow 12k^2 + 4k - 96 = 0 \Rightarrow k = -3$ or $k = \frac{8}{3}$

$\therefore k = -3$

70) b)

$\mathbf{A}^{\mathrm{T}}\mathbf{A} = \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix} = \begin{pmatrix} k & 4 \\ 2k & 6k \end{pmatrix} \begin{pmatrix} k & 2k \\ 4 & 6k \end{pmatrix} = 32 \begin{pmatrix} 1 & 4 \\ 4 & 20 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} k^2 + 16 & 2k^2 + 24k \\ 2k^2 + 24k & 40k^2 \end{pmatrix} = \begin{pmatrix} 32 & 128 \\ 128 & 640 \end{pmatrix}$

$k^2 + 16 = 32 \Rightarrow k^2 = 16 \Rightarrow k = \pm 4$

$2k^2 + 24k = 128 \Rightarrow 2k^2 + 24k - 128 = 0 \Rightarrow k = -16 \text{ or } k = 4$

$\therefore k = 4$

71) a)

$\mathbf{A}\mathbf{A}^{\mathrm{T}}\mathbf{A} = \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 0 \\ k & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3k^2 + 2 & k^3 + 4k & k^2 + 3 \\ k^3 + 3k & 3k^2 + 1 & 2k \\ 1 & k & 2 \end{pmatrix}$

71) b)

$\mathbf{A}^{\mathrm{T}}\mathbf{A}\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}^{\mathrm{T}}$

$= \begin{pmatrix} 1 & k & 0 \\ k & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 1 \\ k & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & k & 0 \\ k & 1 & 0 \\ 1 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 3k^2 + 2 & k^3 + 3k & 1 \\ k^3 + 4k & 3k^2 + 1 & k \\ k^2 + 3 & 2k & 2 \end{pmatrix}$

72)

$(\mathbf{AB})^{\mathrm{T}} = \left(\begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix} \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix} \right)^{\mathrm{T}} = \begin{pmatrix} 16 & 11 \\ 2 & 4 \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} 16 & 2 \\ 11 & 4 \end{pmatrix}$

$\mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} 5 & 3 \\ 1 & 2 \end{pmatrix}^{\mathrm{T}} \begin{pmatrix} 3 & 1 \\ 0 & 2 \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} 5 & 1 \\ 3 & 2 \end{pmatrix} \begin{pmatrix} 3 & 0 \\ 1 & 2 \end{pmatrix} = \begin{pmatrix} 16 & 2 \\ 11 & 4 \end{pmatrix}$

$\therefore (\mathbf{AB})^{\mathrm{T}} = \mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}$

73)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$(\mathbf{AB})^{\mathrm{T}} = \mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}} \Rightarrow \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} 2 & 1 \\ 3 & 1 \end{pmatrix}\right)^{\mathrm{T}} = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2a+3b & a+b \\ 2c+3d & c+d \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$

$\Rightarrow \begin{pmatrix} 2a+3b & 2c+3d \\ a+b & c+d \end{pmatrix} = \begin{pmatrix} 12 & 5 \\ 7 & 3 \end{pmatrix}$

$2a + 3b = 12$ (1)

$a + b = 7$ (2)

Solve (1) and (2) to get $a = 9$ and $b = -2$

$2c + 3d = 5$ (3)

$c + d = 3$ (4)

Solve (3) and (4) to get $c = 4$ and $d = -1$

$\therefore \mathbf{A} = \begin{pmatrix} 9 & -2 \\ 4 & -1 \end{pmatrix}$

74)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$

$(\mathbf{AB})^{\mathrm{T}} = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\begin{pmatrix} e & f \\ g & h \end{pmatrix}\right)^{\mathrm{T}} = \begin{pmatrix} ae+bg & af+bh \\ ce+dg & cf+dh \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} ae+bg & ce+dg \\ af+bh & cf+dh \end{pmatrix}$

$\mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}^{\mathrm{T}}\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} e & g \\ f & h \end{pmatrix}\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} ae+bg & ce+dg \\ af+bh & cf+dh \end{pmatrix}$

$\therefore (\mathbf{AB})^{\mathrm{T}} \equiv \mathbf{B}^{\mathrm{T}}\mathbf{A}^{\mathrm{T}}$ for 2 × 2 matrices

75)

$(\mathbf{AB}^{\mathrm{T}} - \mathbf{BA}^{\mathrm{T}})^2 = -25\mathbf{I}$

$$\Rightarrow \left(\begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix}\begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix}^{\mathrm{T}} - \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix}\begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix}^{\mathrm{T}}\right)^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \left(\begin{pmatrix} x & 2 \\ 1 & y \end{pmatrix}\begin{pmatrix} 3 & 2y \\ x & 1 \end{pmatrix} - \begin{pmatrix} 3 & x \\ 2y & 1 \end{pmatrix}\begin{pmatrix} x & 1 \\ 2 & y \end{pmatrix}\right)^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \left(\begin{pmatrix} 5x & 2xy+2 \\ xy+3 & 3y \end{pmatrix} - \begin{pmatrix} 5x & xy+3 \\ 2xy+2 & 3y \end{pmatrix}\right)^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} 0 & xy-1 \\ 1-xy & 0 \end{pmatrix}^2 = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} (xy-1)(1-xy) & 0 \\ 0 & (xy-1)(1-xy) \end{pmatrix} = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -(xy-1)(xy-1) & 0 \\ 0 & -(xy-1)(xy-1) \end{pmatrix} = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$$\Rightarrow \begin{pmatrix} -(xy-1)^2 & 0 \\ 0 & -(xy-1)^2 \end{pmatrix} = \begin{pmatrix} -25 & 0 \\ 0 & -25 \end{pmatrix}$$

$-(xy-1)^2 = -25 \Rightarrow (xy-1)^2 = 25 \Rightarrow xy - 1 = \pm 5 \Rightarrow xy = 1 \pm 5$

x and y are both positive, so we must have that $xy = 6$

x and y are both integers greater than 1, and $x < y$, so $x = 2$ and $y = 3$

76)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$ and $\mathbf{B} = \begin{pmatrix} e & f \\ g & h \end{pmatrix}$

$$(\mathbf{A}+\mathbf{B})^{\mathrm{T}} = \left(\begin{pmatrix} a & b \\ c & d \end{pmatrix} + \begin{pmatrix} e & f \\ g & h \end{pmatrix}\right)^{\mathrm{T}} = \begin{pmatrix} a+e & b+f \\ c+g & d+h \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} a+e & c+g \\ b+f & d+h \end{pmatrix}$$

$$\mathbf{A}^{\mathrm{T}} + \mathbf{B}^{\mathrm{T}} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}^{\mathrm{T}} + \begin{pmatrix} e & f \\ g & h \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix} + \begin{pmatrix} e & g \\ f & h \end{pmatrix} = \begin{pmatrix} a+e & c+g \\ b+f & d+h \end{pmatrix}$$

$\therefore (\mathbf{A}+\mathbf{B})^{\mathrm{T}} \equiv \mathbf{A}^{\mathrm{T}} + \mathbf{B}^{\mathrm{T}}$ for 2 × 2 matrices

77)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$$(k\mathbf{A})^{\mathrm{T}} = \left(k\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right)^{\mathrm{T}} = \begin{pmatrix} ka & kb \\ kc & kd \end{pmatrix}^{\mathrm{T}} = \begin{pmatrix} ka & kc \\ kb & kd \end{pmatrix}$$

$$k\mathbf{A}^{\mathrm{T}} = k\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{\mathrm{T}} = k\begin{pmatrix} a & c \\ b & d \end{pmatrix} = \begin{pmatrix} ka & kc \\ kb & kd \end{pmatrix}$$

$\therefore (k\mathbf{A})^{\mathrm{T}} \equiv k\mathbf{A}^{\mathrm{T}}$ for 2 × 2 matrices

78)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\det \mathbf{A} = a \times d - b \times c = ad - bc$

$\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$

$\det(\mathbf{A}^{\mathrm{T}}) = a \times d - c \times b = ad - bc$

$\therefore \det(\mathbf{A}^{\mathrm{T}}) \equiv \det \mathbf{A}$ for 2×2 matrices

79)

Let $\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$

$\mathbf{A}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{b}{ad-bc} \\ -\frac{c}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$

$(\mathbf{A}^{-1})^{\mathrm{T}} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{c}{ad-bc} \\ -\frac{b}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$

$\mathbf{A}^{\mathrm{T}} = \begin{pmatrix} a & c \\ b & d \end{pmatrix}$

$(\mathbf{A}^{\mathrm{T}})^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -c \\ -b & a \end{pmatrix} = \begin{pmatrix} \frac{d}{ad-bc} & -\frac{c}{ad-bc} \\ -\frac{b}{ad-bc} & \frac{a}{ad-bc} \end{pmatrix}$

$\therefore (\mathbf{A}^{-1})^{\mathrm{T}} \equiv (\mathbf{A}^{\mathrm{T}})^{-1}$ for 2×2 matrices